Quadratic Functions

Three Forms · Transformations · Graphing · Analysis

Part 1 Three Forms of a Quadratic Function

Every quadratic function can be written in three different but equivalent forms. Each form highlights different features of the parabola.

Form	Equation	What it tells you
Vertex Form	$y = a(x-h)^2 + k$	Vertex $= (h,\, k)$; direction; stretch
Standard Form	$y = ax^2 + bx + c$	y-intercept $= (0, c)$; axis of symmetry
Intercept Form	$y = a(x-x_1)(x-x_2)$	x-intercepts (zeros) $= x_1$ and x_2

Key Idea. In all three forms, the sign of a tells you the direction of the parabola: $a > 0$ opens **up**; $a < 0$ opens **down**. The value of $|a|$ controls width: $|a| > 1$ is narrower than $y = x^2$; $0 < |a| < 1$ is wider.

Example 1. Identify the form and key features of each function.

(a) $y = 2(x-3)^2 + 1$ Vertex form. Vertex $= (3, 1)$, opens up, narrower than $y = x^2$.

(b) $y = x^2 - 5x + 6$ Standard form. y-intercept $= (0, 6)$.

(c) $y = -2(x-1)(x+4)$ Intercept form. Zeros at $x = 1$ and $x = -4$, opens down.

Part 2 Transformations of $f(x) = x^2$

A **transformation** changes the position, size, or orientation of the parent parabola $y = f(x) = x^2$. There are four main types.

1. Translation (Shift)

A translation moves the parabola without changing its shape.

Transformation	Equation	Effect on graph
Shift right h units	$g(x) = f(x-h) = (x-h)^2$	Vertex moves right
Shift left h units	$g(x) = f(x+h) = (x+h)^2$	Vertex moves left
Shift up k units	$g(x) = f(x) + k = x^2 + k$	Vertex moves up
Shift down k units	$g(x) = f(x) - k = x^2 - k$	Vertex moves down

Important: Inside the parentheses, the sign is opposite to the direction. $f(x+2)$ shifts *left* 2; $f(x-2)$ shifts *right* 2.

Example 2. Start with $f(x) = x^2$.

$g(x) = (x-3)^2 + 4$: shift **right 3**, then **up 4**. Vertex moves from $(0,0)$ to $(3,4)$.

$g(x) = (x+2)^2 - 1$: shift **left 2**, then **down 1**. Vertex moves to $(-2,-1)$.

2. Vertical Stretch and Shrink

Multiplying $f(x)$ by a positive constant a changes the *width*.

$\lvert a\rvert > 1$ (e.g., $y = 3x^2$)	Vertical **stretch** — parabola is **narrower**
$0 < \lvert a\rvert < 1$ (e.g., $y = \frac{1}{2}x^2$)	Vertical **shrink** — parabola is **wider**

Every y-value is multiplied by a. The point $(1,1)$ on $y = x^2$ becomes $(1,a)$ on $y = ax^2$.

Example 3.

$y = 2x^2$: vertical stretch by factor 2 — **narrower** than $y = x^2$.

$y = \frac{1}{2}x^2$: vertical shrink by factor $\frac{1}{2}$ — **wider** than $y = x^2$.

3. Reflection

Type	**Equation**	**Effect**
Reflect over x-axis	$g(x) = -f(x) = -x^2$	Flips upside down (opens down)
Reflect over y-axis	$g(x) = f(-x) = (-x)^2 = x^2$	For $y = x^2$, no visible change

For a general function, reflection over the y-axis replaces x with $-x$. For asymmetric quadratics like $y = (x-3)^2$, it matters: $g(x) = f(-x) = (-x-3)^2 = (x+3)^2$ — the vertex moves from $(3,0)$ to $(-3,0)$.

Example 4. Start with $y = x^2$.

Step 1 Reflect over x-axis: $y = -x^2$.

Step 2 Vertical stretch by 2: $y = -2x^2$.

Step 3 Shift right 5: $y = -2(x-5)^2$.

Step 4 Shift up 8: $y = -2(x-5)^2 + 8$. Vertex $(5,8)$.

4. Summary Table — All Transformations

Starting from $y = f(x) = x^2$:

New function	Transformation
$g(x) = f(x) + k$	Shift up k units
$g(x) = f(x) - k$	Shift down k units
$g(x) = f(x - h)$	Shift right h units
$g(x) = f(x + h)$	Shift left h units
$g(x) = a\,f(x),\ a > 1$	Vertical stretch (narrower)
$g(x) = a\,f(x),\ 0 < a < 1$	Vertical shrink (wider)
$g(x) = -f(x)$	Reflection over x-axis
$g(x) = f(-x)$	Reflection over y-axis

Part 3 Writing Equations from a Graph

When you see a graph, pick the form that matches the information given.

Strategy A — Use intercept form when you can read the x-intercepts x_1, x_2:

$$y = a(x - x_1)(x - x_2)$$

Then plug in one other known point to solve for a.

Strategy B — Use vertex form when you can read the vertex (h, k):

$$y = a(x - h)^2 + k$$

Then plug in one other known point to solve for a.

Example 5. A parabola has zeros at $x = 3$ and $x = 7$, and passes through $(5, 8)$.

Step 1 Write intercept form: $y = a(x - 3)(x - 7)$.

Step 2 Plug in $(5, 8)$: $\quad 8 = a(5 - 3)(5 - 7) = a \cdot 2 \cdot (-2) = -4a$.

Step 3 Solve: $a = -2$.

Result: $y = -2(x - 3)(x - 7)$

Standard form: $y = -2(x^2 - 10x + 21) = -2x^2 + 20x - 42$.

Part 4 Converting to Vertex Form

Given $y = ax^2 + bx + c$ in standard form, find the vertex (h, k):

$$\boxed{h = -\frac{b}{2a}} \qquad \boxed{k = f(h)}$$

Then rewrite as $y = a(x - h)^2 + k$.

Example 6. Convert $y = -2x^2 + 20x - 42$ to vertex form.

Step 1 Find h: $h = -\dfrac{20}{2(-2)} = -\dfrac{20}{-4} = 5$.

Step 2 Find k: $k = f(5) = -2(25) + 20(5) - 42 = -50 + 100 - 42 = 8$.

Result: $y = -2(x-5)^2 + 8$. Vertex $(5, 8)$.

Example 7 (Completing the Square). Convert $y = 2x^2 - 8x + 8$.

$$\begin{aligned} y &= 2(x^2 - 4x) + 8 \\ &= 2(x^2 - 4x + 4 - 4) + 8 \\ &= 2(x-2)^2 - 8 + 8 \\ &= 2(x-2)^2 \end{aligned}$$

Vertex $(2, 0)$. The parabola touches the x-axis at its vertex.

Part 5 Domain, Range, and Zeros

For any quadratic $y = ax^2 + bx + c$:

Feature	**Rule**
Domain	Always $(-\infty, +\infty)$
Range ($a > 0$, opens up)	$[k, +\infty)$ where k is the minimum
Range ($a < 0$, opens down)	$(-\infty, k]$ where k is the maximum
Zeros (x-intercepts)	Set $y = 0$ and solve

Example 8. $h(x) = 16x^2 - 4$.

Vertex form: $h(x) = 16(x-0)^2 - 4$, so vertex $= (0, -4)$, opens up.

Zeros: $16x^2 - 4 = 0 \Rightarrow x^2 = \frac{1}{4} \Rightarrow x = \pm\frac{1}{2}$.

Domain: $(-\infty, +\infty)$. **Range:** $[-4, +\infty)$.

Example 9. $y = -(x+5)(x-1)$.

Zeros: $x = -5$ and $x = 1$.

Vertex x-coordinate: midpoint of zeros $= \dfrac{-5+1}{2} = -2$.

$y(-2) = -(-2+5)(-2-1) = -(3)(-3) = 9$. Vertex $= (-2, 9)$. Opens down.

Domain: $(-\infty, +\infty)$. **Range:** $(-\infty, 9]$.

Practice Problems

30 problems covering all five topics above.

Section A Three Forms (Problems 1–5)

1. Identify the form of each function and state the indicated feature.

(a) $y = -3(x-4)^2 + 7$ Form: _____ Vertex: _____

(b) $y = x^2 - 7x + 10$ Form: _____ y-intercept: _____

(c) $y = 5(x+2)(x-6)$ Form: _____ Zeros: _____

2. Write the function in **standard form**: $y = (x-3)(x+5)$.

3. Write the function in **standard form**: $y = -2(x-1)(x-6)$.

4. Write a **vertex form** equation for a parabola with vertex $(-2, 5)$ that passes through $(0, 1)$. Find the value of a first.

5. A quadratic has y-intercept $(0, 12)$ and zeros at $x = 3$ and $x = 4$. Write the function in intercept form, find a, then convert to standard form.

Section B Translations (Problems 6–10)

6. Starting with $f(x) = x^2$, write the equation after shifting **right 4 units** and **up 3 units**. State the new vertex.

7. Starting with $f(x) = x^2$, write the equation after shifting **left 5 units** and **down 2 units**.

8. Describe the transformations that take $f(x) = x^2$ to $g(x) = (x+3)^2 - 7$. State the vertex.

9. The graph of $g(x) = f(x-2)+6$ is a translation of $f(x) = x^2$. (a) Describe the translation. (b) Write the equation of $g(x)$. (c) State the vertex.

10. Point $(1, 1)$ is on $y = x^2$. After the translation $g(x) = (x-4)^2 + 2$, what are the new coordinates of this point?

Section C Vertical Stretch and Shrink (Problems 11–14)

11. Compare each function to $y = x^2$. State whether the parabola is *narrower*, *wider*, or the *same width*, and by what factor.

(a) $y = 5x^2$ (b) $y = \frac{1}{4}x^2$ (c) $y = 0.8\,x^2$

12. Starting with $f(x) = x^2$, write the equation after a **vertical stretch by factor 3**, then a **shift up 5 units**.

13. Starting with $f(x) = x^2$, write the equation after a **vertical shrink by** $\frac{1}{2}$, then a **shift right 3 units** and **down 1 unit**.

14. The point $(2, 4)$ is on $y = x^2$. What is the corresponding point on $y = \frac{3}{4}x^2$? What about on $y = 4x^2$?

Section D Reflections (Problems 15–17)

15. Reflect $f(x) = 2(x - 3)^2 + 1$ over the x**-axis**. Write the new equation and state the new vertex and direction.

16. Reflect $f(x) = (x - 4)^2$ over the y**-axis**. Write the new equation and state the new vertex.

17. Starting with $y = x^2$: first reflect over the x-axis, then shift left 3, then up 5. Write the final equation and identify the vertex.

Section E Combined Transformations (Problems 18–22)

18. List ALL transformations applied to $f(x) = x^2$ to produce $g(x) = -4(x+1)^2 - 3$. State them in the correct order.

19. List ALL transformations applied to $f(x) = x^2$ to produce $p(x) = \frac{1}{2}(x-2)^2 + 4$.

20. Write the equation obtained by applying these steps to $y = x^2$ in order:

(1) Reflect over x-axis (2) Stretch vertically by factor 2 (3) Shift right 1 unit (4) Shift down 4 units

21. The function $g(x) = -\frac{1}{2}(x+3)^2 + 5$ is compared to $f(x) = x^2$. Describe all differences in direction, width, and position.

22. Write the vertex form equation of a parabola that is: *reflected over the x-axis, vertically stretched by 3, shifted right 2, shifted up 6.* What is the vertex?

Section F Writing Equations from Graphs (Problems 23–26)

23. A parabola opens upward, has x-intercepts at $x = -1$ and $x = 5$, and passes through $(2, -9)$. Write the equation in intercept form, then convert to standard form.

24. A parabola has vertex $(3, -2)$ and passes through $(5, 6)$. Write in vertex form, then standard form.

25. A parabola has x-intercepts $(-4, 0)$ and $(2, 0)$ and vertex $(-1,\ 9)$. Find a, write the intercept form, then expand to standard form.

26. From a graph: the parabola opens downward with vertex $(1, 4)$ and y-intercept $(0, 3)$. Find a, write the vertex form, and state the domain and range.

Section G Converting to Vertex Form (Problems 27–28)

27. Convert each function to vertex form using the vertex formula $h = -\dfrac{b}{2a},\ k = f(h)$.

(a) $y = x^2 - 6x + 5$

(b) $y = 3x^2 + 12x - 1$

(c) $y = -x^2 + 4x + 12$

28. Convert $y = 2x^2 - 12x + 10$ to vertex form by **completing the square** (show all steps). Then state the vertex.

Section H Domain, Range, and Zeros (Problems 29–30)

29. For each function, find: **(i)** the zeros, **(ii)** the vertex, **(iii)** the domain, **(iv)** the range.

(a) $f(x) = 2x^2 - 8x + 8$

(b) $y = -(x + 5)(x - 1)$

(c) $h(x) = 16x^2 - 4$

30. A quadratic function opens downward, has zeros at $x = -3$ and $x = 7$, and a maximum value of 20.

(a) Find the x-coordinate of the vertex (midpoint of zeros).

(b) Write the vertex form equation.

(c) Find a.

(d) State the range.

Answer Key

1. (a) Vertex form; vertex $(4, 7)$
(b) Standard form; y-int $(0, 10)$
(c) Intercept form; zeros $x = -2,\ 6$

2. $y = (x-3)(x+5) = x^2 + 2x - 15$

3. $-2(x-1)(x-6) = -2(x^2 - 7x + 6)$
$= -2x^2 + 14x - 12$

4. $y = a(x+2)^2 + 5$. Plug in $(0, 1)$:
$1 = a(4) + 5 \Rightarrow a = -1$.
$y = -(x+2)^2 + 5$

5. $y = a(x-3)(x-4)$. Plug in $(0, 12)$:
$12 = a(3)(4) = 12a \Rightarrow a = 1$.
$y = (x-3)(x-4) = x^2 - 7x + 12$

6. $g(x) = (x-4)^2 + 3$; vertex $(4, 3)$

7. $g(x) = (x+5)^2 - 2$; vertex $(-5, -2)$

8. Left 3, down 7; vertex $(-3, -7)$

9. (a) Right 2, up 6
(b) $g(x) = (x-2)^2 + 6$
(c) Vertex $(2, 6)$

10. $(1, 1)$ shifts to $(1+4,\ 1+2) = (5, 3)$

11. (a) $a = 5 > 1$: narrower, factor 5
(b) $a = \frac{1}{4} < 1$: wider, factor $\frac{1}{4}$
(c) $a = 0.8 < 1$: wider, factor 0.8

12. $g(x) = 3x^2 + 5$

13. $g(x) = \frac{1}{2}(x-3)^2 - 1$

14. On $\frac{3}{4}x^2$: $(2,\ 3)$. On $4x^2$: $(2, 16)$

15. $g(x) = -2(x-3)^2 - 1$;
vertex $(3, -1)$, opens down

16. $g(x) = (-x-4)^2 = (x+4)^2$;
vertex $(-4, 0)$

17. $g(x) = -(x+3)^2 + 5$; vertex $(-3, 5)$

18. (1) Vertical stretch $\times 4$
(2) Reflect over x-axis
(3) Shift left 1
(4) Shift down 3

19. (1) Vertical shrink $\times \frac{1}{2}$
(2) Shift right 2
(3) Shift up 4

20. $g(x) = -2(x-1)^2 - 4$

21. Opens down ($a < 0$); wider ($|a| = \frac{1}{2}$); shifted left 3 and up 5; vertex $(-3, 5)$

22. $g(x) = -3(x-2)^2 + 6$; vertex $(2, 6)$

23. $y = a(x+1)(x-5)$. Plug in $(2, -9)$:
$-9 = a(3)(-3) = -9a \Rightarrow a = 1$.
$y = (x+1)(x-5) = x^2 - 4x - 5$

24. $y = a(x-3)^2 - 2$. Plug in $(5, 6)$:
$6 = a(4) - 2 \Rightarrow a = 2$.
$y = 2(x-3)^2 - 2 = 2x^2 - 12x + 16$

25. Vertex $x = \frac{-4+2}{2} = -1$.
$y = a(x+4)(x-2)$. Plug in $(-1, 9)$:
$9 = a(3)(-3) = -9a \Rightarrow a = -1$.
$y = -(x+4)(x-2) = -x^2 - 2x + 8$

26. $y = a(x-1)^2 + 4$. Plug in $(0, 3)$:
$3 = a(1) + 4 \Rightarrow a = -1$.
$y = -(x-1)^2 + 4$
Domain: $(-\infty, +\infty)$; Range: $(-\infty, 4]$

27a. $h = -\frac{-6}{2} = 3$, $k = 9 - 18 + 5 = -4$.
$y = (x-3)^2 - 4$

27b. $h = -\frac{12}{6} = -2$, $k = 12 - 24 - 1 = -13$.
$y = 3(x+2)^2 - 13$

27c. $h = -\frac{4}{-2} = 2$, $k = -4 + 8 + 12 = 16$.
$y = -(x-2)^2 + 16$

28.

$$\begin{aligned} y &= 2(x^2 - 6x) + 10 \\ &= 2(x^2 - 6x + 9 - 9) + 10 \\ &= 2(x-3)^2 - 18 + 10 \\ &= 2(x-3)^2 - 8 \end{aligned}$$

Vertex $(3, -8)$

29a. $f(x) = 2(x-2)^2$; zero: $x = 2$;
vertex $(2, 0)$; domain $(-\infty, +\infty)$; range $[0, +\infty)$

29b. Zeros $x = -5, 1$; vertex $(-2, 9)$;
domain $(-\infty, +\infty)$; range $(-\infty, 9]$

29c. Zeros $x = \pm\frac{1}{2}$; vertex $(0, -4)$;
domain $(-\infty, +\infty)$; range $[-4, +\infty)$

30. (a) $x = \frac{-3+7}{2} = 2$
(b) $y = a(x-2)^2 + 20$
(c) Use zero $x = 7$: $0 = a(25) + 20 \Rightarrow a = -\frac{4}{5}$.
$y = -\frac{4}{5}(x-2)^2 + 20$
(d) Range: $(-\infty, 20]$

End of Quadratic Functions Notes & Practice